

HirePrep AI: An AI–Powered Resume Analyzer and Skill Gap Suggestions

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PROBLEM STATEMENT

- To develop an AI/ML-based system that analyzes a candidate's resume against a specific job role to identify skill gaps and provide personalized suggestions for improving job readiness.

MOTIVATION

- Many job seekers submit resumes without tailoring them to the job role, leading to poor ATS (Applicant Tracking System) compatibility.
- Recruiters receive thousands of resumes, making it difficult to manually filter them for skill match.
- AI and ML can automate and personalize the resume screening and enhancement process.
- Helps job seekers become more competitive by understanding exactly what skills they lack.

OBJECTIVE

- To build a system that reads and parses resumes using NLP.
- To extract and match skills from resumes against the selected job role.
- To calculate ATS score based on keyword relevance and formatting.
- To identify skill gaps and suggest skills the candidate should acquire.
- To provide a user-friendly interface for inputting resumes and job roles.

SCOPE

- Focuses on resume parsing (PDF/DOCX format).
- Supports multiple job roles and industries.
- Uses AI/ML to compare resume skills with job requirements.
- Can be extended to suggest online courses or certifications.
- Can integrate with job portals or LinkedIn APIs (optional enhancement).

LIST OF REQUIRED HARDWARE, SOFTWARE, OR OTHER EQUIPMENT

- Software Requirements

Operating System: Windows 11 / Ubuntu Linux

- Programming Languages

Python (NLP, ML Models)

JavaScript/TypeScript (Frontend if web app in Angular/React)

- Database: MySQL / MongoDB

- Version Control: Git + GitHub

- IDE/Tools: VS Code / PyCharm, Postman (API Testing), Jupyter Notebook

- Frameworks & Libraries (Python):

Flask / FastAPI (Backend API)

spaCy, NLTK (Resume Parsing)

scikit-learn (TF-IDF, Cosine Similarity)

Hugging Face Transformers (BERT embeddings)

Pandas, NumPy (Data Handling)

Frontend (if web app): Angular / React / Streamlit

SYSTEM OVERVIEW

- Resume Parser – Extracts content from resumes.
- Skill Extractor – Identifies and classifies relevant skills.
- Job Role Matcher – Compares resume skills with job role skills.
- Skill Gap Analyzer – Finds missing or weak areas.
- Recommendation Engine – Suggests skills or learning resources.
- Frontend Interface – Allows users to upload and view results.

EXPECTED OUTCOMES

- A web-based tool or application for users to upload resumes and get analysis.
- Automated ATS score and feedback on job fit.
- List of missing skills with suggestions.
- Improvement roadmap for job seekers.

ARCHITECTURE AND INITIAL PHASE OF DESIGN (DFD).

